

HACK4UCAR 2025

Problem Statement

University of Carthage (UCAR) · Digitalization & Institutional Intelligence

1. Context & Problem Statement

The University of Carthage (UCAR) oversees more than 30 affiliated institutions, each operating with fragmented, non-digitalized systems. This operational reality generates major inefficiencies at every level of governance.

Current State — Key Findings

- Data scattered across multiple formats (paper, Excel files, disconnected tools)
- No centralized system for data management
- Slow and inefficient decision-making processes
- Complex and unreliable performance monitoring across institutions

Observed Impacts

- Poor visibility over operational and academic data
- Redundant work across teams and institutions
- Limited coordination between affiliated institutions
- Inability to monitor overall performance in real time

2. Your Mission

Design and develop AI-powered solutions to help UCAR:

- Centralize and structure operational, academic, and financial data
- Automate repetitive administrative processes
- Enable smarter, data-driven strategic decision-making
- Improve communication and coordination across institutions

Target Vision

Build an intelligent university management platform enabling real-time centralization of operational, academic, financial, and environmental data, in service of strategic decision-making driven by key performance indicators (KPIs).

3. Expected Deliverables

Participants must deliver:

- **A working prototype**
 - Web and/or mobile application, scalable to 30+ institutions
- **A clear and demonstrated AI integration**
 - AI must be at the core of the solution, not just a surface-level UI element
- **A short pitch (5 to 7 minutes)**
 - Explaining impact, feasibility, and scalability

4. Judging Criteria

Criterion	Description
Impact	Real usefulness for UCAR, addressing genuine institutional needs
Innovation	Quality and depth of AI integration in the solution
Usability	Ease of use for non-technical staff
Scalability	Ability to handle 30+ institutions without degradation
Feasibility	Realism of deployment in a Tunisian university context

5. Competition Tracks

Track	Scope & Examples
 Track 1 — Data Digitalization & Structuring	◦ Convert unstructured data (PDFs, images, Excel) into usable databases ◦ OCR + AI for intelligent data extraction ◦ Automated ingestion and validation pipelines
 Track 2 — Smart Analytics & Decision Support	◦ Dashboards with multi-level AI insights ◦ Trend prediction (budget, staffing, resources) ◦ Anomaly detection and automated alerts
 Track 3 — AI Assistants & Automation	◦ Chatbots for querying institutional data in natural language ◦ Automated periodic report generation ◦ NLP interfaces for non-technical staff
 Track 4 — End-to-End Smart Platform	◦ Combine multiple features into one unified system ◦ Consolidated multi-institution view with KPI aggregation ◦ Predictive analytics engine + automated reports + intelligent alerts

Important Note — Key Requirements from UCAR

The following elements represent the priority expectations expressed by the UCAR team. They must be taken into account in the design and evaluation of all solutions.

► Consolidated Multi-Institution View

- Aggregation and comparison of KPIs across all 30+ institutions in real time
- Intelligent alerts on detected anomalies or exceeded critical thresholds
- Automated periodic reports (weekly, monthly, annual)
- Predictive analytics engine to anticipate institutional trends and risks

► Dashboard Covering All Institutional Processes

The solution must cover all university processes with their own indicators and synthesis reports:

- Enrollment — Exams — Pedagogy
- Strategy — Partnerships — Student Life
- Finance — Human Resources — Training
- Research — Infrastructure — Equipment
- Inventory — ESG/CSR — Logistics

6. Key Performance Indicators (KPIs) — Reference Framework

Solutions should ideally integrate (or partially cover) the following KPI families, defined by UCAR as priority management indicators:

Domain	Key Indicators (non-exhaustive examples)
 Academic	Success rate • Attendance rate • Grade repetition rate • Dropout rate • Pedagogical progression • Exam results
 Employment	Employability rate • Time to employment post-graduation • National partnership rate • International partnership rate
 Finance	Allocated vs consumed budget • Cost per student • Budget execution rate • Spending by department
 ESG / CSR	Energy consumption • Carbon footprint • Recycling rate • Sustainable mobility • Campus accessibility
 HR	Teaching / administrative headcount • Absenteeism rate • Training completed • Teaching load • Team stability
 Research	Number of publications • Active projects • Funding secured • Academic partnerships • Patents filed
 Infrastructure	Classroom occupancy rate • IT equipment status • Equipment availability • Ongoing works
 Partnerships	Active agreements • Incoming/outgoing student mobility • International projects • Academic networks

7. Technical Recommendations

To maximize your impact, the following technical directions are strongly recommended:

- **Multi-tenant architecture**
 - A single deployment capable of managing all institutions in an isolated and secure manner
- **Unified data model**
 - A common schema enabling cross-institution comparison and aggregation

- **Explainable AI pipeline**
 - Alerts and predictions must be understandable by non-data-scientists
- **No-code / low-code interface for administrators**
 - Presidents and deans do not code — UX must be flawless and intuitive
- **Automatically exportable PDF/Excel reports**
 - Integration with existing processes without forcing a complete operational break

Good luck to all teams! 🚀

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